

STAT 1210

Chapter 2 Homework

Fall 2026

Questions

Question 1

A freshwater researcher measured the dissolved oxygen concentration (mg/L) in a pond on ten mornings:

7.8 8.1 7.6 8.3 8.0 7.9 7.5 8.0 7.7 7.8

1. Calculate the mean dissolved oxygen concentration.
2. Interpret the mean in the context of the study.
3. Is the mean necessarily one of the observed measurements? Explain.

Question 2

The waiting times, in minutes, for eleven clients at a community service office were

18 22 25 27 31 34 50 20 22 48 34

1. Find the median waiting time and state its location in the ordered list.
2. An eighth client with a waiting time of 29 minutes is added. Find the new median.
3. Why does the method for finding the median change when the sample size changes from odd to even?

Question 3

A transit agency recorded the delays, in minutes, for five buses on a route:

2 3 3 4 5

A data-entry error changes the last value from 5 minutes to 35 minutes.

1. Calculate the mean and median before the error.
2. Calculate the mean and median after the error.
3. Which measure of centre is more resistant to this unusual value?
4. If the agency reported only the mean after the error, what misleading impression might be created?

Question 4

Eight water samples have a reported mean nitrate concentration of 6.4 mg/L. Seven of the measurements are

5.8 6.1 6.3 6.5 6.7 6.8 6.2

1. What total concentration is required for all eight measurements?
2. Find the missing measurement.
3. Verify your answer by recalculating the mean.
4. Why is it important to use all eight observations when converting the mean back to a total?

Question 5

For each distribution below, decide whether the mean is likely to be less than, approximately equal to or greater than the median. Then choose either the mean and standard deviation or the median and interquartile range as the more appropriate summaries.

1. Annual household repair costs, where most households have modest costs but a few require very expensive repairs.
2. Repeated readings from a calibrated instrument that form an approximately symmetric distribution with no outliers.
3. Scores on an easy examination, where most students score highly but a small number receive much lower scores.

Explain each choice using the shape of the distribution.

Question 6

Biologists counted frog calls during evening surveys at two wetlands. Wetland A was surveyed 12 times and had a mean of 18 calls per survey. Wetland B was surveyed 28 times and had a mean of 11 calls per survey.

1. How many frog calls were recorded across the surveys at each wetland?
2. Calculate the mean number of calls per survey across all 40 surveys.
3. A student averages the two reported means and obtains 14.5. Why is this not the correct combined mean?
4. Under what condition would simply averaging the two group means produce the correct combined mean?

Question 7

The operating lives, in days, of 11 air-quality sensors are listed in increasing order:

42 45 47 48 50 52 54 55 58 60 81

Use the hand-calculation method from class: when n is odd, exclude the overall median before finding the quartiles.

1. Find the median.
2. Find Q_1 and Q_3 .
3. Calculate the IQR.
4. Write the five number summary.
5. Interpret the IQR in context.

Question 8

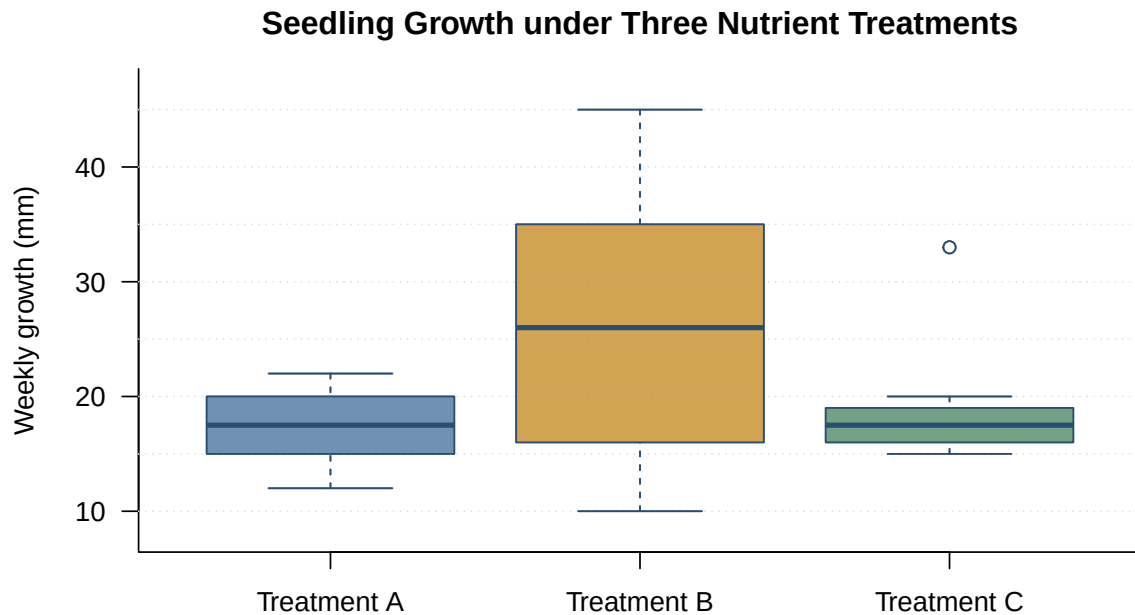
The commute times, in minutes, for 12 workers are

18 20 22 24 25 27 29 31 33 35 38 44

1. Find the median.
2. Find Q_1 and Q_3 using the median-of-halves method.
3. Calculate the IQR and write the five number summary.
4. Approximately what percentage of observations lie between Q_1 and Q_3 ?
5. Does the IQR describe the full spread of the commute times? Explain.

Question 9

A greenhouse compared the weekly growth of seedlings receiving three nutrient treatments. The boxplots summarize the illustrative results.



1. Which treatment has the largest median weekly growth?
2. Which treatment has the largest IQR?
3. Which treatment has the smallest IQR, and what does that indicate?
4. Identify the suspected outlier and describe it in context.
5. Why would comparing only the full ranges give an incomplete picture of consistency?

Question 10

The processing times, in days, for 12 permit applications were

2 3 4 4 5 6 6 7 8 9 10 18

1. Calculate the five number summary.
2. Calculate the IQR.
3. Calculate the lower and upper outlier fences using the $1.5 \times IQR$ rule.
4. Is any observation a suspected outlier?
5. Describe the boxplot that should be drawn, including the locations of the box, median, whiskers and any separate point.

Question 11

A county monitors nitrate concentrations in groundwater. For one set of readings, the first and third quartiles are $Q_1 = 2.4$ mg/L and $Q_3 = 3.2$ mg/L. The observed concentrations are

1.9 2.1 2.4 2.6 2.8 3.0 3.2 3.5 4.6 7.8

1. Calculate the IQR.
2. Calculate the lower and upper outlier fences.
3. Which readings are identified as suspected outliers?
4. Does the $1.5 \times IQR$ rule prove that these values are errors? Explain.
5. Give two reasonable actions the county should take before deciding whether to remove or correct either value.

Question 12

Seven deliveries initially had the following travel times in minutes:

20 22 23 24 26 27 28

Suppose the final delivery instead takes 70 minutes because of a major road closure.

1. Calculate the mean and median for the original data.
2. Calculate the mean and median after replacing 28 with 70.
3. Calculate the IQR before and after the replacement.
4. Which numerical summaries are resistant to this change?
5. Without calculating first, would you expect the standard deviation to increase, decrease or remain unchanged? Explain. Then use the values in the solution to check your reasoning.

Question 13

The operating times, in hours, of five prototype batteries are

4 6 8 10 12

1. Calculate the sample mean.
2. Find each deviation $x_i - \bar{x}$ and verify that the deviations sum to 0.
3. Calculate the sum of squared deviations.
4. Calculate the sample variance using

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

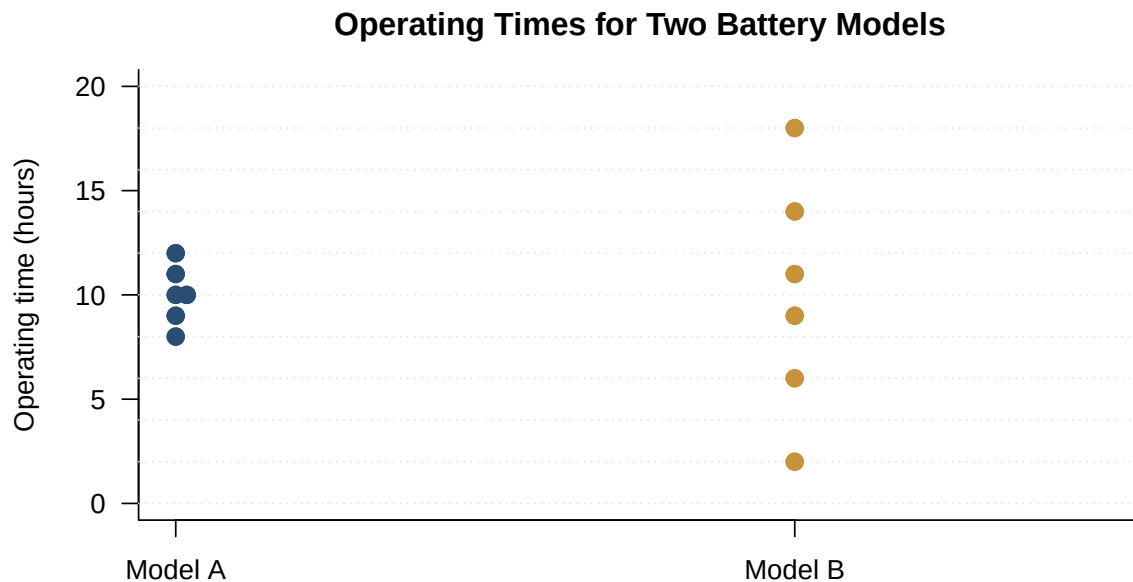
5. Calculate and interpret the sample standard deviation.
6. Why is the standard deviation easier to interpret than the variance in this example?

Question 14

Two prototype battery models have the following operating times.

Model A: 8, 9, 10, 10, 11, 12

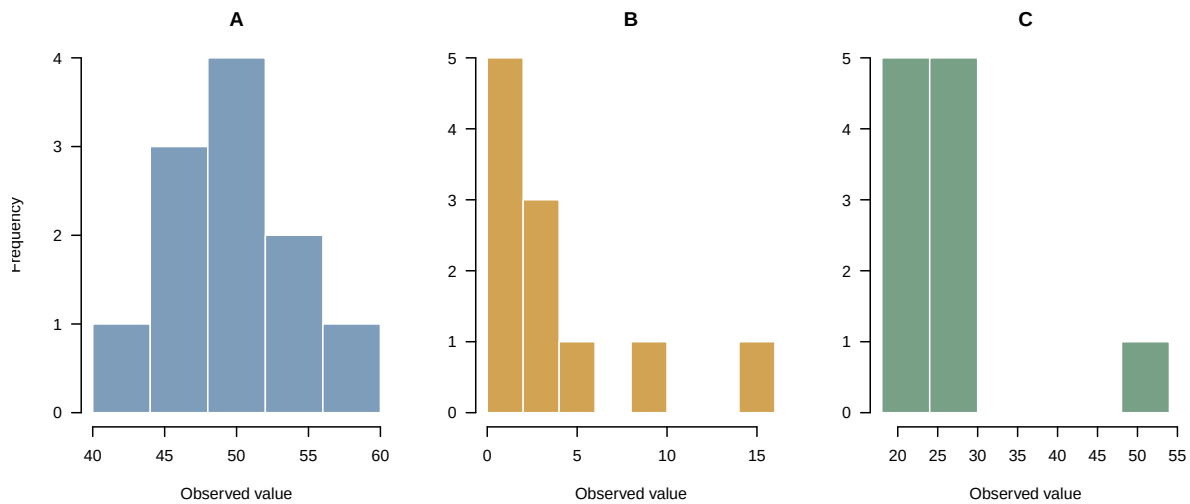
Model B: 2, 6, 9, 11, 14, 18



1. Calculate the mean and median for each model.
2. Which model has the larger range?
3. Calculate the sample standard deviation for each model.
4. Which model is more consistent in its operating time? Explain.
5. Why would reporting only the means hide an important difference between the models?

Question 15

The histograms below show three illustrative distributions. The units differ across the three settings, so compare their shapes rather than their numerical scales.



For each histogram:

1. Describe the shape and identify any visible unusual feature.
2. State whether the mean is likely to be less than, approximately equal to or greater than the median.
3. Choose the mean and standard deviation or the median and IQR as the more appropriate summaries.
4. Explain why the same pair of summaries is not automatically best for every distribution.

Question 16

A news report describes the following sale prices, in thousands of dollars, for ten homes in a small neighbourhood:

240 250 255 260 265 270 280 295 310 1200

1. Calculate the mean and median sale price.
2. Calculate Q_1 , Q_3 and the IQR using the hand-calculation method.
3. Use the $1.5 \times IQR$ rule to determine whether the 1200 value is a suspected outlier.
4. The report calls the mean of \$362500 the “typical home price”. Why could that statement be misleading?
5. Give a more informative numerical summary for a prospective buyer.

Question 17

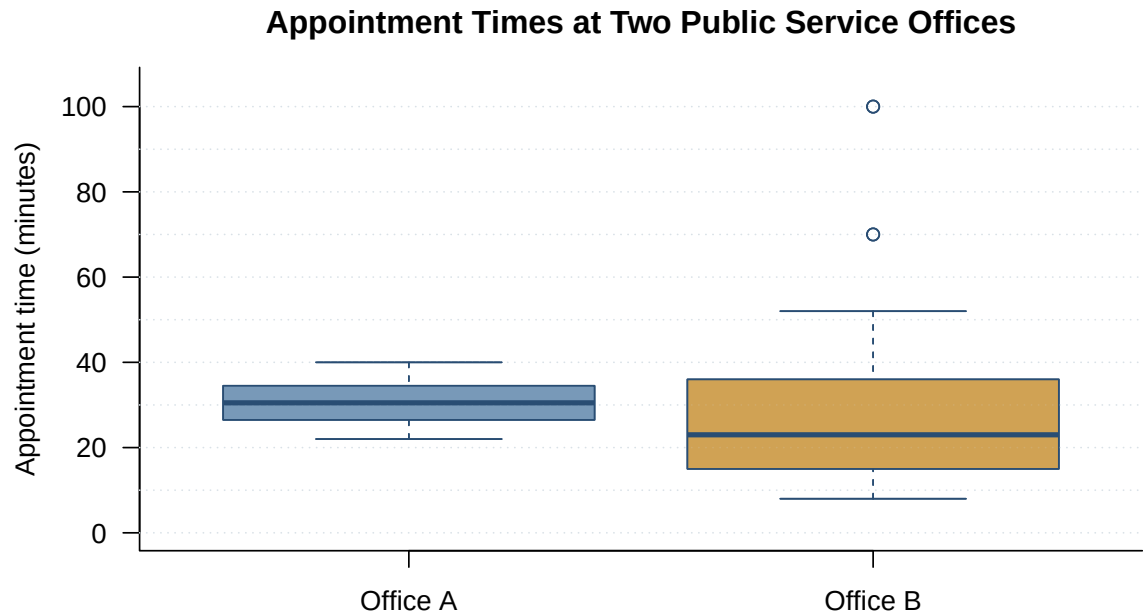
A student enters the following lake-temperature measurements into R:

```
lake_temperature <- c(14.2, 14.8, 15.1, 15.4, 15.6, 16.0, 16.2, 16.5, 17.1, 21.8)
```

1. Write one-line R commands to calculate the mean, median, five-number summary, IQR and sample standard deviation.
2. Use the output reported in the solution to compare the mean and median.
3. What feature of the data explains the difference between these two measures?
4. Which measures of centre and spread would you report?
5. What should be investigated before deciding how to handle the largest measurement?

Question 18

Two public service offices recorded the time required to complete the same type of appointment.



The numerical summaries are:

Office	Mean	Median	Standard deviation	IQR
A	30.63	30.5	5.19	7.5
B	30.75	23.0	24.63	18.5

1. The two means are nearly identical. Why do the distributions nevertheless describe very different service experiences?
2. Which office has the larger typical appointment time if the median is used?
3. Which office has more consistent appointment times? Support your answer with two numerical summaries.
4. Which pair of centre and spread measures is more suitable for Office A? Which pair is more suitable for Office B?
5. What do the boxplot and summaries suggest the manager of Office B should investigate?
6. Write a short, neutral comparison of the two offices that reports both centre and variability without claiming that one office caused the differences.

Solutions

Solution to Question 1

1. The mean is

$$\bar{x} = \frac{7.8 + 8.1 + 7.6 + 8.3 + 8.0 + 7.9 + 7.5 + 8.0 + 7.7 + 7.8}{10} = \frac{78.7}{10} = 7.87 \text{ mg/L}$$

2. Across the six mornings, the pond had an average dissolved oxygen concentration of 7.87 mg/L.
3. No. The mean is a balance point calculated from all the observations, so it does not have to equal any individual measurement. In this dataset, 7.87 mg/L was not observed.

The numerical calculations can be checked in R:

```
oxygen <- c(7.8, 8.1, 7.6, 8.3, 8.0, 7.9, 7.5, 8.0, 7.7, 7.8)
mean(oxygen)
```

```
## [1] 7.87
```

Solution to Question 2

1. For $n = 11$, the median is in position

$$\frac{n+1}{2} = \frac{12}{2} = 6$$

The ordered data is

18, 20, 22, 22, 25, 27, 31, 34, 34, 48, 50

The sixth observation is 27 so the median is 27 minutes.

2. The ordered data become

18, 20, 22, 22, 25, 27, 29, 31, 34, 34, 48, 50

. For $n = 12$, the median is the average of the 6th and 7th observations:

$$M = \frac{27 + 29}{2} = 28 \text{ minutes}$$

3. With an odd number of observations, one observation lies at the exact centre. With an even number, the centre falls between the two middle observations, so their average is used.

The numerical calculations can be checked in R.

```
waiting_times <- c(18, 22, 25, 27, 31, 34, 50, 20, 22, 48, 34)
median(waiting_times)
```

```
## [1] 27
```

```
new_waiting_times <- c(18, 22, 25, 27, 31, 34, 50, 20, 22, 48, 34, 29)
median(new_waiting_times)
```

```
## [1] 28
```

Solution to Question 3

1. Before the error

$$\bar{x} = \frac{2 + 3 + 3 + 4 + 5}{5} = 3.4 \text{ minutes}$$

and the median is 3 minutes.

2. After the error

$$\bar{x} = \frac{2 + 3 + 3 + 4 + 35}{5} = 9.4 \text{ minutes}$$

while the median remains 3 minutes.

3. The median is more resistant. The extreme value increases the mean by 6 minutes but does not change the middle observation.
4. The mean of 9.4 minutes could suggest that a typical bus was delayed by almost 10 minutes even though four of the five recorded delays were between 2 and 4 minutes. The error should be checked before any summary is reported.

The numerical calculations can be checked in R:

```
original_delays <- c(2, 3, 3, 4, 5)
mean(original_delays)
```

```
## [1] 3.4
```

```
median(original_delays)
```

```
## [1] 3
```

```
delays_with_error <- c(2, 3, 3, 4, 35)
mean(delays_with_error)
```

```
## [1] 9.4
```

```
median(delays_with_error)
```

```
## [1] 3
```

Solution to Question 4

1. Because

$$\bar{x} = \frac{\sum_{i=1}^{n=8} x_i}{8}$$

the required total is

$$\sum_{i=1}^{n=8} x_i = \bar{x} \times 8 = (6.4)(8) = 51.2 \text{ mg/L}$$

2. The seven known values sum to

$$\bar{x} = \frac{\sum_{i=1}^{n=7} x_i}{7} = 44.4 \text{ mg/L}$$

So the missing value is

$$51.2 - 44.4 = 6.8 \text{ mg/L}$$

3. Including the missing value gives

$$\frac{44.4 + 6.8}{8} = \frac{51.2}{8} = 6.4 \text{ mg/L}$$

which agrees with the reported mean.

4. The reported mean describes eight samples. Multiplying by 7 would reconstruct the total for the wrong number of observations and could not produce the correct missing value.

The numerical calculations can be checked in R.

```
known_nitrate <- c(5.8, 6.1, 6.3, 6.5, 6.7, 6.8, 6.2)
required_total <- 8 * 6.4
missing_value <- required_total - sum(known_nitrate)
print(missing_value)
```

```
## [1] 6.8
```

```
mean(c(known_nitrate, missing_value))
```

```
## [1] 6.4
```

Solution to Question 5

1. Repair costs are likely right skewed, so the mean is likely greater than the median. The median and IQR are preferable because they resist the influence of the few very large costs.
2. For a reasonably symmetric distribution with no outliers, the mean and median should be approximately equal. The mean and standard deviation are appropriate because the centre and spread can be described without strong distortion from skewness or outliers.
3. The examination scores are likely left skewed, so the mean is likely less than the median. The median and IQR are preferable because the lower tail can pull the mean downward and inflate the standard deviation.

Solution to Question 6

1. Wetland A contributed $(12)(18) = 216$ calls and Wetland B contributed $(28)(11) = 308$ calls.
2. The combined mean is
$$\bar{x}_{\text{combined}} = \frac{216 + 308}{12 + 28} = \frac{524}{40} = 13.1 \text{ calls per survey}$$
3. The value 14.5 gives the two wetlands equal weight even though Wetland B supplied more than twice as many surveys. A combined mean must weight each group mean by its group size.
4. A simple average of the group means would be correct if the two groups had equal sample sizes.

The numerical calculations can be checked in R.

```
calls_a <- 12 * 18
calls_b <- 28 * 11
```

```
calls_a
```

```
## [1] 216
```

```
calls_b
```

```
## [1] 308
```

```
(calls_a + calls_b) / (12 + 28)
```

```
## [1] 13.1
```

Solution to Question 7

1. The sixth observation is the median, so $M = 52$ days.
2. After excluding the median, the lower half is 42, 45, 47, 48, 50, whose median is 47. The upper half is 54, 55, 58, 60, 81, whose median is 58. Therefore, $Q_1 = 47$ and $Q_3 = 58$ days.

3.

$$IQR = Q_3 - Q_1 = 58 - 47 = 11 \text{ days}$$

4. The five number summary is

$$(\min = 42, Q_1 = 47, Q_2 = 52, Q_3 = 58, \max = 81)$$

5. The middle 50% of sensor lifetimes span 11 days, from 47 to 58 days.

The numerical calculations can be checked in R:

```
sensor_life <- c(42, 45, 47, 48, 50, 52, 54, 55, 58, 60, 81)
summary(sensor_life)
```

```
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
##    42.00   47.50   52.00   53.82   56.50   81.00
```

Solution to Question 8

1.

$$M = \frac{27 + 29}{2} = 28 \text{ minutes}$$

2. The lower half is 18, 20, 22, 24, 25, 27 so

$$Q_1 = \frac{22 + 24}{2} = 23$$

The upper half is 29, 31, 33, 35, 38, 44 so

$$Q_3 = \frac{33 + 35}{2} = 34$$

3.

$$IQR = 34 - 23 = 11 \text{ minutes}$$

and the five-number summary is

$$(\min = 18, Q_1 = 23, Q_2 = 28, Q_3 = 34, \max = 44)$$

4. Approximately 50% of the observations lie between Q_1 and Q_3 .

5. No. The IQR measures the spread of the middle 50% and deliberately ignores the most extreme quarters of the data. The full range is $44 - 18 = 26$ minutes.

The numerical calculations can be checked in R:

```
commute_times <- c(18, 20, 22, 24, 25, 27, 29, 31, 33, 35, 38, 44)
summary(commute_times)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    18.00   23.50   28.00   28.83   33.50   44.00
```

Solution to Question 9

1. Treatment B has the largest median, approximately 26 mm.
2. Treatment B has the largest IQR, so its middle 50% is the most spread out.
3. Treatment C has the smallest IQR. Its central half of seedling growth values is the most tightly clustered, although the unusual high value must also be considered.
4. The value of 33 mm under Treatment C is shown as a suspected high outlier. One seedling grew substantially more than the others receiving that treatment.
5. The range uses only the minimum and maximum and is highly sensitive to an outlier. Treatment C has a fairly large full range because of one unusual seedling even though its middle 50% is tightly clustered. The IQR and the complete boxplot give a more informative comparison.

Solution to Question 10

1. The five-number summary is

$$(\min = 2, Q_1 = 4, Q_2 = 6, Q_3 = 8.5, \max = 18)$$

- 2.

$$IQR = 8.5 - 4 = 4.5 \text{ days}$$

3. The fences are

$$Q_1 - 1.5(IQR) = 4 - 1.5(4.5) = -2.75$$

and

$$Q_3 + 1.5(IQR) = 8.5 + 1.5(4.5) = 15.25$$

4. The observation of 18 days lies above the upper fence so it is a suspected outlier.
5. The box extends from 4 to 8.5 days with a median line at 6. The lower whisker reaches 2, the upper whisker reaches the largest value of 10 and 18 is plotted separately.

The numerical calculations can be checked in R.

```
permit_times <- c(2, 3, 4, 4, 5, 6, 6, 7, 8, 9, 10, 18)
permit_times_summary <- summary(permit_times)
print(permit_times_summary)
```

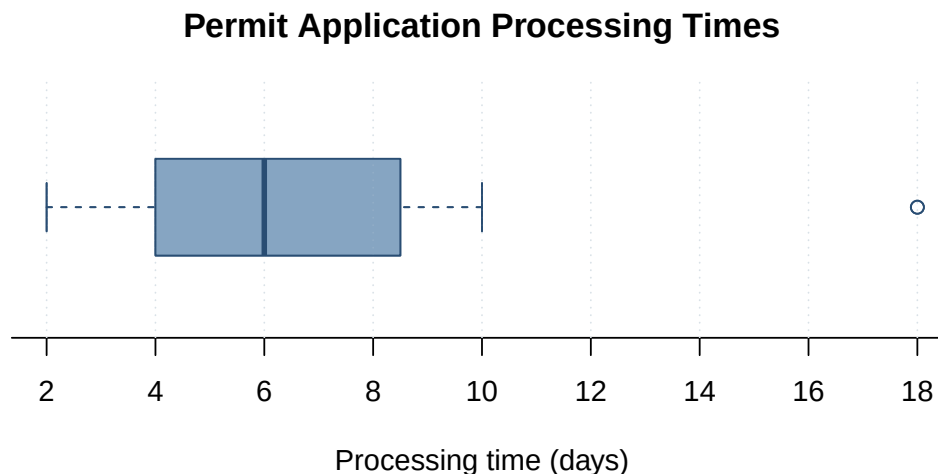
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.000   4.000   6.000   6.833   8.250  18.000
```

```
q1 <- permit_times_summary[[2]]
q3 <- permit_times_summary[[5]]
iqr <- q3 - q1

upper_limit <- q1 + 1.5 * iqr
lower_limit <- q1 - 1.5 * iqr

permit_times[permit_times < lower_limit | permit_times > upper_limit]
```

```
## [1] 18
```



Solution to Question 11

1.

$$IQR = 3.2 - 2.4 = 0.8 \text{ mg/L}$$

2. The lower fence is

$$2.4 - 1.5(0.8) = 1.2 \text{ mg/L}$$

and the upper fence is

$$3.2 + 1.5(0.8) = 4.4 \text{ mg/L}$$

3. The readings 4.6 mg/L and 7.8 mg/L are above the upper fence and are therefore suspected outliers. No reading is below the lower fence.
4. No. The rule identifies observations that deserve attention; it does not determine their cause. The values could be accurate evidence of local contamination, or they could result from sampling, recording or equipment problems.
5. The county could verify the original records and instrument calibration and it could collect repeat samples from the same locations. Information about location, weather and nearby land use could also help explain the readings.

The numerical calculations can be checked in R:

```
readings <- c(1.9, 2.1, 2.4, 2.6, 2.8, 3.0, 3.2, 3.5, 4.6, 7.8)
q1 <- 2.4
q3 <- 3.2
iqr <- q3 - q1
fences <- c(lower = q1 - 1.5 * iqr, upper = q3 + 1.5 * iqr)

iqr
```

```
## [1] 0.8
```

```
fences
```

```
## lower upper
##  1.2    4.4
```

```
readings[readings < fences["lower"] | readings > fences["upper"]]
```

```
## [1] 4.6 7.8
```

Solution to Question 12

1. For the original data

$$\bar{x} = \frac{20 + 22 + 23 + 24 + 26 + 27 + 28}{7} \approx 24.29 \text{ minutes}$$

and the median is 24 minutes.

2. After the replacement

$$\bar{x} = \frac{20 + 22 + 23 + 24 + 26 + 27 + 70}{7} \approx 30.29 \text{ minutes}$$

while the median remains 24 minutes.

3. Originally, $Q_1 = 22$ and $Q_3 = 27$, so $IQR = 5$ minutes. After the replacement, the lower and upper halves have the same middle values, so $Q_1 = 22$, $Q_3 = 27$ and $IQR = 5$ minutes again.
4. The median and IQR are resistant here. The mean is not resistant and increases by 6 minutes.
5. The standard deviation should increase because 70 is much farther from the mean than the original value of 28. The sample standard deviation increases from approximately 2.87 minutes to 17.67 minutes.

Solution to Question 13

1. The mean is

$$\bar{x} = \frac{4 + 6 + 8 + 10 + 12}{5} = 8 \text{ hours}$$

2. The deviations are

$$x_1 - \bar{x} = 4 - 8 = -4$$

$$x_2 - \bar{x} = 6 - 8 = -2$$

$$x_3 - \bar{x} = 8 - 8 = 0$$

$$x_4 - \bar{x} = 10 - 8 = 2$$

$$x_5 - \bar{x} = 12 - 8 = 4$$

and their sum is

$$-4 - 2 + 0 + 2 + 4 = 0$$

3. The squared deviations are

$$(x_1 - \bar{x})^2 = (4 - 8)^2 = (-4)^2 = 16$$

$$(x_2 - \bar{x})^2 = (6 - 8)^2 = (-2)^2 = 4$$

$$(x_3 - \bar{x})^2 = (8 - 8)^2 = (0)^2 = 0$$

$$(x_4 - \bar{x})^2 = (10 - 8)^2 = (2)^2 = 4$$

$$(x_5 - \bar{x})^2 = (12 - 8)^2 = (4)^2 = 16$$

so their sum is $16 + 4 + 0 + 4 + 16 = 40$.

- 4.

$$s^2 = \frac{40}{5 - 1} = 10 \text{ hours}^2$$

- 5.

$$s = \sqrt{10} \approx 3.16 \text{ hours}$$

A battery lifetime is typically about 3.16 hours away from the sample mean of 8 hours.

6. The standard deviation is measured in hours, the same units as the data. The variance is measured in squared hours, which has a less direct practical interpretation.

The numerical calculations can be checked in R.

```
battery_life <- c(4, 6, 8, 10, 12)
mean(battery_life)
```

```
## [1] 8
```

```
deviations <- battery_life - mean(battery_life)
```

```
deviations
```

```
## [1] -4 -2  0  2  4
```

```
sum(deviations)
```

```
## [1] 0
```

```
deviations^2
```

```
## [1] 16  4  0  4 16
```

```
sum(deviations^2)
```

```
## [1] 40
```

```
var(battery_life)
```

```
## [1] 10
```

```
sd(battery_life)
```

```
## [1] 3.162278
```

Solution to Question 14

1. Both models have a mean of 10 hours and a median of 10 hours.
2. Model A has range $12 - 8 = 4$ hours. Model B has range $18 - 2 = 16$ hours, so Model B has the larger range.
3. For Model A, $s \approx 1.41$ hours. For Model B, $s \approx 5.69$ hours.
4. Model A is more consistent because its observations are much more tightly clustered. Both its range and standard deviation are smaller.
5. The equal means describe the same centre but not the very different spreads. A user who values predictable operating time would need a measure of variation as well as the mean.

The numerical calculations can be checked in R:

```
model_a <- c(8, 9, 10, 10, 11, 12)
model_b <- c(2, 6, 9, 11, 14, 18)

c(mean = mean(model_a), median = median(model_a),
  range = diff(range(model_a)), sd = sd(model_a))
```

```
##      mean  median   range      sd
## 10.000000 10.000000  4.000000  1.414214
```

```
c(mean = mean(model_b), median = median(model_b),
  range = diff(range(model_b)), sd = sd(model_b))
```

```
##      mean  median   range      sd
## 10.0000  10.0000 16.0000  5.6921
```

Solution to Question 15

1. Distribution A is approximately symmetric with no clear outlier. Distribution B is right-skewed with a long upper tail. Distribution C has a compact main group and one unusually large observation.
2. For A, the mean and median should be approximately equal. For B, the right tail should pull the mean above the median. For C, the large observation should also pull the mean above the median.
3. The mean and standard deviation are reasonable for A. The median and IQR are preferable for B and C because they are resistant to skewness and the unusually large observations.
4. The usefulness of a summary depends on distribution shape. The mean and standard deviation use every value and can be strongly affected by a long tail or outlier, while the median and IQR give a more stable description in those settings.

Solution to Question 16

1. The values sum to 3625 thousand dollars, so

$$\bar{x} = \frac{3625}{10} = 362.5 \text{ thousand dollars}$$

The median is

$$M = \frac{265 + 270}{2} = 267.5 \text{ thousand dollars}$$

2. The lower half is 240, 250, 255, 260, 265, giving $Q_1 = 255$. The upper half is 270, 280, 295, 310, 1200, giving $Q_3 = 295$. Therefore,

$$IQR = 295 - 255 = 40 \text{ thousand dollars}$$

3. The upper fence is

$$295 + 1.5(40) = 355 \text{ thousand dollars.}$$

Since $1200 > 355$, the \$1200000 sale is a suspected high outlier.

4. The unusually expensive home pulls the mean far above nine of the ten prices. The mean therefore does not represent the centre of the main group very well.
5. A more informative report would give the median price of \$267500 together with $Q_1 = \$255000$, $Q_3 = \$295000$ and an IQR of \$40000. It should also mention the unusually expensive sale rather than silently discarding it.

Solution to Question 17

1. The requested commands are shown below.

```
mean(lake_temperature)
```

```
## [1] 16.27
```

```
median(lake_temperature)
```

```
## [1] 15.8
```

```
summary(lake_temperature)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    14.20   15.18   15.80   16.27   16.43   21.80
```

```
IQR(lake_temperature)
```

```
## [1] 1.25
```

```
sd(lake_temperature)
```

```
## [1] 2.11925
```

2. R gives a mean of 16.27°C and a median of 15.80°C . The mean is higher.
3. The measurement of 21.8°C lies well above the other observations and pulls the mean upward. The distribution is right-skewed because of this large value.
4. The median and IQR are the more resistant summaries. Using R's default quartile calculation, $Q_1 = 15.175^{\circ}\text{C}$, $Q_3 = 16.425^{\circ}\text{C}$ and $IQR = 1.25^{\circ}\text{C}$.
5. The student should check the original record, time and location of measurement, instrument operation and field conditions. The value may be an error but it may also describe a real local event.

Solution to Question 18

1. Office A has a compact, roughly symmetric distribution. Office B has many shorter appointments and a small number of very long appointments. Those long appointments raise Office B's mean until it is close to Office A's mean, even though its median and spread are quite different.
2. Office A has the larger median: 30.5 minutes compared with 23.0 minutes for Office B.
3. Office A is more consistent. Its standard deviation is 5.19 minutes rather than 24.63 minutes, and its IQR is 7.5 minutes rather than 18.5 minutes.
4. For Office A, the mean and standard deviation are reasonable because the distribution is fairly symmetric and has no clear extreme observation. For Office B, the median and IQR are more suitable because the distribution is strongly right-skewed and contains very long appointment times.
5. The manager should investigate the unusually long appointments, including whether certain appointment types, staffing conditions, system interruptions or data-recording issues explain them. The values should not be deleted merely because they are unusual.
6. A neutral comparison could state: The offices had similar mean appointment times, about 31 minutes. Office A had a higher median time but substantially less variation. Office B had a lower median of 23 minutes, together with a right-skewed distribution and several much longer appointments.

The numerical calculations can be checked in R:

```
c(mean = mean(office_a), median = median(office_a), sd = sd(office_a), IQR = IQR(office_a))
```

```
##      mean   median      sd      IQR
## 30.625000 30.500000  5.188127  7.500000
```

```
c(mean = mean(office_b), median = median(office_b), sd = sd(office_b), IQR = IQR(office_b))
```

```
##      mean   median      sd      IQR
## 30.750000 23.000000 24.63466 18.500000
```